

1. Administrative & Study Identification

Full Study Title: A Study Evaluating the Efficacy and Safety of Multiple Immunotherapy-Based Treatment Combinations in Patients With Advanced Liver Cancers (Morpheus-Liver) **Official Title:** A Phase Ib/II, Open-Label, Multicenter, Randomized Umbrella Study Evaluating the Efficacy and Safety of Multiple Immunotherapy-Based Treatment Combinations in Patients With Advanced Liver Cancers (Morpheus-Liver) **Short Title/Acronym:** MORPHEUS-LIVER **Protocol Number:** G042216 **NCT Number:** NCT04524871 **Posting ID:** 419156380615963440 **Sponsor/Company Name:** Roche **Investigational Product:** Atezolizumab + Bevacizumab (Trade names: Avastin, Vegzelma, Alymsys, Zirabev, Mvasi) combined with various experimental agents (e.g., NKT2152, R07247669, and others) **Phase of Development:** Phase I/Phase II (PHASE1, PHASE2) **Trial Status:** RECRUITING **Medical Monitor:** Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of multiple immunotherapy-based treatment combinations compared to an active control arm in participants with advanced liver cancers, determined by the investigator-assessed Objective Response Rate (ORR) on two consecutive occasions ≥ 4 weeks apart during Stage 1 per RECIST v1.1. **Secondary Objectives:** To evaluate Progression-Free Survival (PFS), Overall Survival (OS), OS at Specific Timepoints, and to characterize the safety and tolerability profile of the various treatment combinations.

2.2 Background & Rationale This umbrella study is designed with the flexibility to open new treatment arms as new treatments become available, close existing treatment arms that demonstrate minimal clinical activity or unacceptable toxicity, or introduce additional cohorts. Cohort 1 will enroll participants with locally advanced or metastatic hepatocellular carcinoma (HCC) without prior systemic therapy. Participants who experience loss of clinical benefit or unacceptable toxicity during Stage 1 may be eligible to receive treatment with a different treatment combination (Stage 2).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Umbrella / Platform Study) **Design Stage:** - **Stage 1:** Initial randomization to one of several treatment arms. - **Stage 2:** Crossover/sequential assignment for participants experiencing loss of clinical benefit

or unacceptable toxicity during Stage 1. **Blinding:** Open-label
Randomization: Randomized

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 518
participants **Number of Sites:** Multicenter **Trial Status:**
RECRUITING

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria Stage 1: 1. ECOG Performance Status of 0 or 1 within 7 days prior to randomization. 2. Locally advanced or metastatic and/or unresectable HCC with diagnosis confirmed by histology/cytology or clinically by AASLD criteria in cirrhotic patients. 3. Child-Pugh class A within 7 days prior to randomization. 4. Disease not amenable to curative surgical and/or locoregional therapies, and no prior systemic treatment for HCC. 5. Life expectancy $\geq$ 3 months. 6. Availability of a representative tumor specimen suitable for determination of PD-L1 and/or additional biomarker status via central testing.

Stage 1 and Stage 2: 7. Measurable disease according to RECIST v1.1. 8. Adequate hematologic and end-organ function within 7 days prior to initiation of study treatment. 9. Documented virology status of hepatitis, confirmed by screening tests for HBV and HCV; Negative HIV test at screening. 10. Agreement to use contraception/abstinence; men must refrain from donating sperm.

Stage 2: 11. ECOG Performance Status of 0, 1, or 2. 12. Ability to initiate Stage 2 treatment within 3 months after experiencing unacceptable toxicity (not related to atezolizumab or R07247669) or loss of clinical benefit on Stage 1. 13. Availability of a tumor specimen from a biopsy performed upon discontinuation of Stage 1.

NKT2152-Containing Arm Specifics: 14. Total bilirubin $\leq$ 1.5 X ULN in the absence of Gilbert's disease ($\leq$ 3.0 X ULN if Gilbert's disease). 15. AST/ALT $\leq$ 2.5 X ULN ($\leq$ 5 X ULN if liver metastases present).

4.2 Exclusion Criteria Stage 1 Exclusions: 1. Prior treatment with CD137 agonists, immune checkpoint blockade therapies, or inhibitors targeting HIF2a. 2. Investigational therapy or locoregional therapy to the liver within 28 days prior to study initiation. 3. Untreated or incompletely treated esophageal and/or gastric varices with bleeding or high risk for bleeding; prior bleeding event due to varices within 6 months. 4. AEs from prior anti-cancer therapy not resolved to Grade $\leq$ 1 (except alopecia). 5. Inadequately controlled hypertension, history of hypertensive crisis/encephalopathy, or significant vascular disease. 6. Evidence of bleeding diathesis/coagulopathy or history of hemoptysis within 1 month. 7. Core biopsy/minor surgical procedure within 3 days prior. Radiotherapy within 28 days (or abdominal/pelvic radiotherapy within 60 days).

Stage 1 and Stage 2 Exclusions: 8. Fibrolamellar or sarcomatoid HCC, or mixed cholangiocarcinoma and HCC. 9. History of hepatic encephalopathy, moderate or severe ascites. 10. HBV and HCV coinfection. 11. Symptomatic, untreated, or actively progressing CNS metastases, or history of leptomeningeal disease. 12. Active or history of autoimmune disease or immune deficiency. Active TB. 13. Major surgery within 4 weeks. 14. Prior allogeneic stem cell or solid organ transplantation. 15. Severe infection within 4 weeks or therapeutic oral/IV antibiotics within 2 weeks prior to initiation. 16. Treatment with a live, attenuated vaccine within 4 weeks.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Backbone Agents: Atezolizumab + Bevacizumab (Trade names: Avastin, Vegzelma, Alymsys, Zirabev, Mvasi). **ATC Codes for Bevacizumab:** S01LA08, L01FG01 **Experimental Additions:** Specific cohorts include investigational agents such as NKT2152, R07247669, and other targeted therapies introduced via protocol amendments. **Duration of Treatment:** Administered continuously until unacceptable toxicity or loss of clinical benefit, potentially triggering a transition to Stage 2 regimens.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing disease symptoms. **Prohibited:** Concurrent investigational therapies (within 28 days prior), live vaccines, chronic daily NSAID use, and systemic immunosuppressive medications.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Imaging, Viral screening (HIV, HBV, HCV)
Baseline	Day 1	Randomization, First Dose Administration, Safety Audit
Interim	Every 21 Days	Safety Labs, Immunotherapy Infusions, Ongoing AE monitoring; Imaging evaluations
Stage 2 Transition	Variable	New biopsy (if clinically feasible), assessment for cross-over eligibility
End of Study	Variable	End of Treatment Visit, Final Vital Signs, Exit Evaluation, Long-term Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Outcome Objective Response Rate (ORR): Defined as the proportion of participants with a complete response (CR) or partial response (PR) on two consecutive occasions ≥ 4 weeks apart during Stage 1, as determined by the investigator according to RECIST v1.1.

7.2 Secondary Outcomes 1. Progression-Free Survival (PFS) 2. Overall Survival (OS) 3. OS at Specific Timepoints

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection at each clinical visit, graded per CTCAE criteria. Patients entering Stage 2 must have prior immunotherapy-related AEs resolved to Grade 1 or better. **Serious Adverse Events (SAEs):** Reported continuously per standard oncology reporting timelines.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Efficacy-evaluable population and Safety Set. **Sample Size Justification:** Targeted enrollment of 518 subjects ensures sufficient power to evaluate dynamic treatment arms within the umbrella framework. **Handling of Missing Data:** Standard survival analysis methodologies for PFS and OS endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: The study is conducted in accordance with Good Clinical Practice (GCP) guidelines. **Monitoring Plan:** Routine onsite monitoring interspersed with centralized remote data reviews to verify RECIST endpoints and safety profiles.

11. Study Identifiers & Governance

Sponsor/Company: Roche **Posting ID:** 419156380615963440 **Primary ID:** G042216 **Registry Number:** NCT04524871 **MeSH Code:** D008110 **SNOMED ID:** 421042008

12. Clinical Framework (Disease Specifics)

Disease Area: Advanced Liver Cancers **Preferred UMLS Name:** Advanced Liver Cancer / advanced adult primary liver cancer **Semantic Type:** Neoplastic Process **Diagnosis Threshold:** Locally advanced or metastatic and/or unresectable HCC. Advanced Liver Cancers describe primary liver malignancies extensively disseminated or metastatic, no longer amenable to curative treatments (surgical resection or transplantation).

13. Study Procedures & Methodology

Management Pathway: Advanced HCC standard oncology care pathway.
Education Protocol (HCP): RECIST v1.1 evaluation criteria and comprehensive irAE management training. **Education Protocol (Patient):** Informed consent, reporting of unacceptable toxicity, and adherence to contraception requirements.

14. Observation & Follow-Up Schedule

Total Duration: Until unacceptable toxicity, disease progression, or loss of clinical benefit (with potential crossover to Stage 2). **Onsite Visit Cadence:** Corresponding to treatment cycles.
Checklist Review Interval: Tumor assessment via CT/MRI scans at regular intervals to establish CR/PR on consecutive occasions \$\\ge\$4 weeks apart.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	____	[DD-MMM-YYYY]				